GEN 241 · SHEET 2, PART 2 · LAB WEEK 3

The first true condition chooses the answer

For 70 kg and 175 cm, Q18 gives BMI 22.8571 and the sheet's label Normal. Put the test `bmi < 30` first and the same number prints Overweight. The calculation is right; the decision is wrong.

THIS WEEK'S TASK

Finish Q7, Q9-Q11 and Q13-Q18. Keep last week's eight files in the same `XXX_Sheet_2` folder. The final archive needs all 18.

PROFESSOR'S MESSAGE

Sheet 2 must be completed within two weeks. Counting from the 4 October 2026 message gives **18 October 2026**; this is an inferred planning date. Follow any exact deadline and upload link the professor announces.

بالمصري

اكتب توقّعك الأول، وبعد كده جرّب. ركّز على الرقم اللي عند الحد بالظبط، مش بس المثال السهل. أسماء الملفات وخطوات التسليم بالصور في الفصلين 6 و7.

During the lab

Q7 together, Q18 from your plan, Q14 alone. Vote with your fingers.

After the lab

Use the widgets, then type and test your own programs in Dev-C++. F11 compiles and runs; F9 compiles only. On the TA's Mac, Cmd+Shift+B runs in VS Code.

Worksheet

One practice program. It runs before you edit it. It is not a replacement for the 18 question files.

CHAPTER 1

One test chooses one block

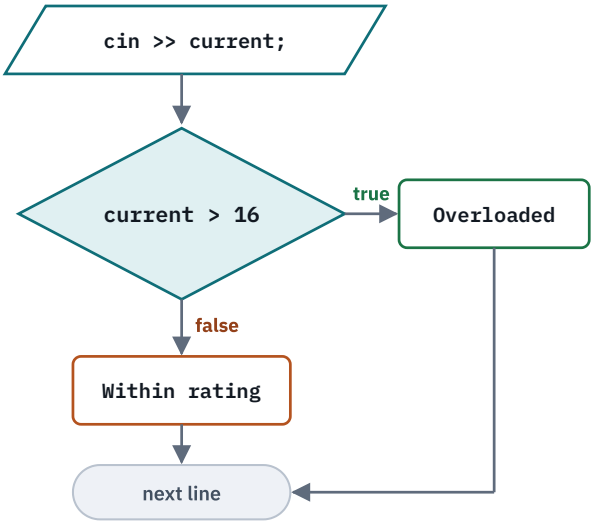
Trace a two-way if/else and explain which statements run.

Read › Decide › Check › Name › Hand in

1.1 Read the test aloud

In this teaching example, a current above 16 A prints Overloaded; otherwise it prints Within rating. This is a simplified comparison exercise, not a model of a real circuit breaker.

`>` means greater than, `<` less than, `>=` and `<=` include equality. Use `==` to compare and `!=` for not equal. One `=` assigns a value.




```
cout << "Current (A): ";
cin >> current;
if (current > 16)
{
    cout << "Overloaded" << endl;
}
else
{
    cout << "Within rating" << endl;
}
```

( ) a test: it is true or false
{ } after if: runs only when true
{ } after else: runs only when false

Follow one path from input to output. Both paths meet after the if/else.

C++ excerpt

```
1  double current;
2  cin >> current;
3  if (current > 16)
4  {
5      cout << "Overloaded" << endl;
6  }
7  else
8  {
9      cout << "Within rating" << endl;
10 }
```

1.2 Braces group statements

The braces make a block. Without braces, an `if` controls only the next statement. A plain `if` can do nothing when false; an `if/else` executes exactly one of its two blocks.

Do not put a semicolon after `if (current > 16)`. That would make an empty statement the controlled statement. A compiler may warn, but it can still produce an executable. Read warnings.

بالمصري

الـ `if` بتسأل سؤال إجابته صح أو غلط. الأقواس `{ }` بتجمع السطور اللي ليها نفس الشرط. الفاصلة المنقوطة بعد الشرط بتخلّيه يتحكم في جملة فاضية.

CHECK YOURSELF

At exactly 16 A, what does the program above print?

A Overloaded

B Within rating

C Both messages

Source: Lecture 3 pp. 10-11 introduces selection and Boolean expressions. Liang §§3.3-3.6 develops the syntax and common errors.

CHAPTER 2

A triangle needs every inequality

Use && for conditions that must all hold, before calculating an area.

Read › **Decide** › **Check** › Name › Hand in

2.1 A safe place for the formula

Q7 asks you to calculate an area only when the three edges form a triangle. Every pair must add to more than the remaining edge. `&&` means all conditions must be true; `||` means at least one; `!` reverses true and false.

Q7 body, after reading double a, b, c; include cmath

```cpp
if (a + b > c && a + c > b && b + c > a)
{
    double s = (a + b + c) / 2;
    double area = pow(s * (s-a) * (s-b) * (s-c), 0.5);
    cout << "Area = " << area << endl;
}
else
{
    cout << "The input is invalid" << endl;
}
```

```
3 + 4 > 5: true
3 + 5 > 4: true
4 + 5 > 3: true
Valid. Area = 6
```

Change the sides. Read each comparison before the combined result.

2.2 Try the case that breaks your rule

`3 4 5` gives area 6. `1 2 3` lies flat, so `>=` would wrongly accept it. `10 2 3` passes `a+b>c`, but fails another pair. One comparison is not enough.

CHECK YOURSELF

Why is the area calculation inside the true block?

A To avoid calculating an area for invalid edges

B Because cout always needs if

بالمصري

قبل ما تحسب المساحة، اتأكد إن الأضلاع ينفع تعمل مثلث. علامة && معناها الشروط كلها، مش شرط واحد بس.

CHAPTER 3

Convert first, then climb the BMI ladder

Turn Q18 into a plan and trace the first matching branch.

Read › **Decide** › **Check** › Name › Hand in

3.1 Five lines before code

Use the labels from the sheet as a programming exercise. Inputs here must be positive.

WANT
: The BMI value in kg/m² and the word from the table.

HAVE
: Weight in kg, height in cm; boundaries 18.5, 25 and 30.

CONVERT
: `heightM = heightCm / 100`. The variable is double, so this is real division.

RELATE
: `bmi = weight / pow(heightM, 2)`, followed by the table from top to bottom.

CHECK
: 70 kg, 175 cm gives 22.8571, Normal. At 76.5625 kg and 175 cm, BMI is exactly 25, Overweight.

Q18 body; include iostream and cmath

```
double weight, heightCm, heightM, bmi;
cin >> weight >> heightCm;
// CONVERT
heightM = heightCm / 100;
// RELATE
bmi = weight / pow(heightM, 2);
cout << "BMI is " << bmi << endl;
if (bmi < 18.5)       cout << "Underweight" << endl;
else if (bmi < 25.0)  cout << "Normal" << endl;
else if (bmi < 30.0)  cout << "Overweight" << endl;
else                  cout << "Obese" << endl;
```

3.2 Watch a compiled run

```
1.  double weight, heightCm, heightM, bmi;
2.  cout << "Weight (kg) and height (cm): ";
3.  cin >> weight >> heightCm;
4.  heightM = heightCm / 100;
5.  bmi = weight / pow(heightM, 2);
6.  if (bmi < 18.5)       cout << "Underweight" << endl;
7.  else if (bmi < 25.0)  cout << "Normal" << endl;
8.  else if (bmi < 30.0)  cout << "Overweight" << endl;
9.  else                  cout << "Obese" << endl;
10. return 0;
```

Step 10 / 10: `return 0;` - finished. To see another branch, run it again with other numbers.

```
weight = 70 | heightCm = 175 | heightM = 1.75 | bmi = 22.8571
```

```
Weight (kg) and height (cm): 70 175
Normal
```

Input: 70 175. These states were recorded from compiled C++, not invented by the browser. The declaration frame means uninitialized, not zero.

3.3 Change one thing

```
Height used = 1.75 m
BMI = 22.8571
Printed label = Normal
```

Keep the inputs fixed to compare a unit error with a branch-order error.

3.4 Why the second test has no lower bound

If execution reaches `bmi < 25`, the first test `bmi < 18.5` already failed. The lower bound is established by the path. Once a branch runs, the later tests are not evaluated.

بالمصري

لو وصلت للشرط التاني، يبقى الأول طلع `false` فعلًا. أول شرط يطلع `true` بينفّذ فرعه، وباقي السلم مش بيتحسب.

CHECK YOURSELF

At BMI exactly 30, which label does the sheet require?

A Overweight

B Obese

CHAPTER 4

Test below, at and above the boundary

Choose inputs that expose a wrong comparison, then plan a new decision.

Read › Decide › Check › Name › Hand in

4.1 Q14 includes the circle edge

Distance from the origin is `pow(x*x + y*y, 0.5)`. The sheet explicitly says less than or equal to 10. So (6,8), whose distance is exactly 10, is in the circle. A reference program may print a separate on-the-circle message; follow the sheet's requested rule here.

C++ excerpt

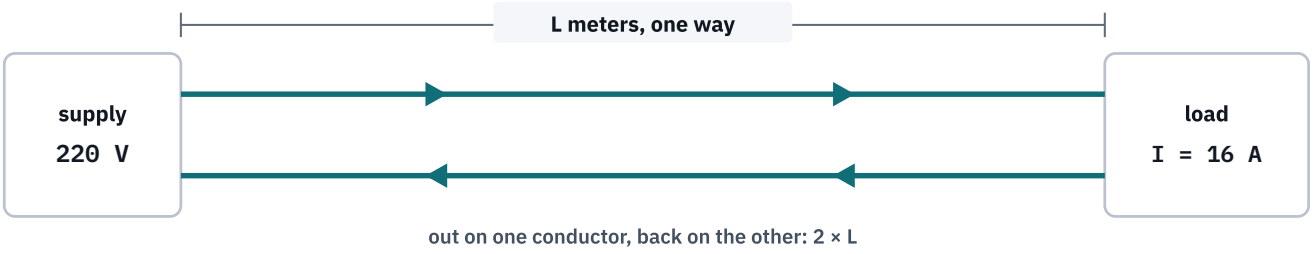
```cpp
double distance = pow(x*x + y*y, 0.5);
if (distance <= 10)
    cout << "in the circle" << endl;
else
    cout << "outside the circle" << endl;
```

```
Distance = 10
Sheet rule <= 10: in the circle
With < 10 instead: outside the circle
```

Compare < with <=. The edge input is what tells them apart.

4.2 Your turn: a cable decision

This extra exercise uses a given resistance of 7.41 Ω/km per conductor, a 16 A current, a 220 V supply and a 5% budget. Treat these as exercise assumptions: two conductors, a resistive load, and reactance ignored.




```
drop = 2 * (L / 1000.0) * 7.41 * 16
```

Given for this exercise: 7.41 Ω per km per conductor

Budget in this exercise: 5% of 220 V. Single-phase, resistive load, reactance ignored.

The current travels out and back. Use twice the one-way length.

In your notebook, write WANT, HAVE, CONVERT, RELATE and CHECK before opening the answer. Read the one-way length L in meters; print the drop in volts and whether it is within the given budget.

– **Compare your plan**

WANT: volts and within/over budget. **HAVE:** L meters and the given constants. **CONVERT:** L/1000.0 km; 0.05*220 = 11 V. **RELATE:** drop = 2*(L/1000.0)*7.41*16; if drop <= 11, within. **CHECK:** 40 m gives 9.4848 V, within; 50 m gives 11.856 V, over.

CHECK YOURSELF

Which test most directly distinguishes < from <= 10 in Q14?

A (4,5)

B (6,8)

C (9,9)

CHAPTER 5

Finish the selection questions

Choose a decision structure and a test for each remaining question.

Read › Decide › Check › Name › Hand in

Lab priority: Q7, Q18, then Q14 alone. Q11 is the extension. Use these routes for the remaining questions; write and test your own files. Keep Q1-Q6, Q8 and Q12 from week 2.

Q9 · three discriminant cases

Assume a is nonzero for a quadratic. Compute d = b*b - 4*a*c. For d > 0, print (-b+sqrt(d))/(2*a) and (-b-sqrt(d))/(2*a); for d == 0, print -b/(2*a); otherwise print exactly "The equation has no real roots." Do not calculate sqrt(d) in the negative case. Use <cmath>.

Test: 1 -3 2 → roots 2 and 1; 1 2 1 → -1; 1 2 5 → no real roots.

Q10 · three separate comparisons

Read integers a, b, c. If a<b, exchange them using temp=a; a=b; b=temp;. Then if a<c, exchange a and c. Finally if b<c, exchange b and c. These are three independent if statements: test all three, although a swap only runs when its test is true.

Test: 1 2 3 → 3 2 1; 3 2 1 → unchanged; 2 2 1 → 2 2 1.

Q11 · validity, then triangle type

Use Q7's validity test first. In the valid branch: if a==b && b==c, Equilateral; else if a==b || a==c || b==c, Isosceles; else Scalene. Test equilateral first because it also satisfies the two-equal test.

Test: 5 5 5 → Equilateral; 5 5 8 → Isosceles; 3 4 5 → Scalene; 1 2 3 → invalid.

Q13 · a valid three-digit integer

Test n>=100 && n<=999. In the valid branch, hundreds=n/100; tens=(n/10)%10; units=n%10; print their sum. Otherwise print invalid.

Test: 456 → 15; 100 → 1; 999 → 27; 99 and 1000 → invalid.

Q15 · check containment before overlap

Find d between the centers using the Q4 distance formula. The supplied hint tests d<=fabs(r1-r2), then d<=r1+r2, then neither. Its first condition detects containment in either direction. To say specifically "circle2 is inside circle1", the directional condition is d+r2<=r1. Ask the TA which interpretation the professor wants before final submission.

Test: 0 0 5 / 0 0 2 → circle2 inside; 0 0 5 / 6 0 2 → overlap; 0 0 5 / 10 0 2 → neither. Reversing the radii exposes the hint's ambiguity.

Q16 · clock arithmetic, then AM/PM

Include <ctime>. Obtain seconds from time(0). second=totalSeconds%60; minute=(totalSeconds/60)%60; hour=(totalSeconds/3600+offset)%24. Read the whole-hour GMT offset. If hour==0, display 12 AM; else if hour<12, display hour AM; else if hour==12, display 12 PM; otherwise display hour-12 PM. Minutes and seconds remain unchanged.

Test: Test the conversion with fixed hour values: 0 → 12 AM; 11 → 11 AM; 12 → 12 PM; 23 → 11 PM. Live seconds change on every run.

Q17 · print the equation in pieces

For x1!=x2, calculate m=(y2-y1)/(x2-x1) and b=y1-m*x1 in double. Start with "y = ". If m!=1, print m; then print "x". If b>0 print " + " and b; else if b<0 print " - " and -b; if b==0 print nothing. The sheet's slope formula assumes a nonvertical line.

Test: 0 3 / 2 5 → y = x + 3; 0 -3 / 1 -1 → y = 2x - 3; 0 0 / 1 1 → y = x.

بالمصري

لو إجابتك مختلفة عن صورة الحل، ارجع لنص السؤال. في Q15 تحديدًا التلميح ممكن يقول إن الدايرة التانية جوه الأولى حتى لو العكس هو الصح. اسأل المعيد قبل ما تسلّم.

CHAPTER 6

Name it, save it: the professor’s rules

After this chapter you can make the sheet folder and every question file with exactly the names the professor specifies, on a lab PC, without help.

Read › Decide › Check › **Name** › **Hand in**

6.1

Five rules to check before submission

Follow the professor's file and folder naming instructions for every sheet. The rules, from the note the professor sent to every student:

1. **No Arabic and no spaces** in any file or folder name. English letters, digits and underscores only.
2. **One folder per sheet**, named `XXX_Sheet_N`. `XXX` is your number on the class list, always three digits.
3. **One file per question**, named `Q_SSQQ.cpp`: two digits for the sheet, two for the question. Underscores, never dashes.

4. **Delete the `.exe` files** before you send. Dev-C++ makes one next to every `.cpp` you run.
5. **Archive the whole folder** into one file with the folder's name, and upload it at the link the professor announces. ZIP and RAR are both allowed; this guide uses Windows ZIP.

Consistent names let the marker identify your class-list number, sheet and question without opening each program. Do not substitute your university ID for your number on the class list.

6.2

Your names, worked out

Type your number on the class list - not your university ID, not the PC's number. The page builds the names the way the rules say.

FOLDER	FILE	ZIP TO SEND
034_Sheet_2	**Q_0201.cpp**	**034_Sheet_2.zip**

Number 34 becomes 034 (always three digits). Sheet 2, question 1 becomes 0201 (two digits each). The full path is Desktop\GEN241\034_Sheet_2\Q_0201.cpp.

The padding is the part people get wrong. Number 7 is `007`, question 1 is `01`. The names are computed by this page from the rules, as you type.

6.3

Check a name before you trust it

Type any file or folder name, or press one of the examples. Each one is a mistake that really happens in the lab.

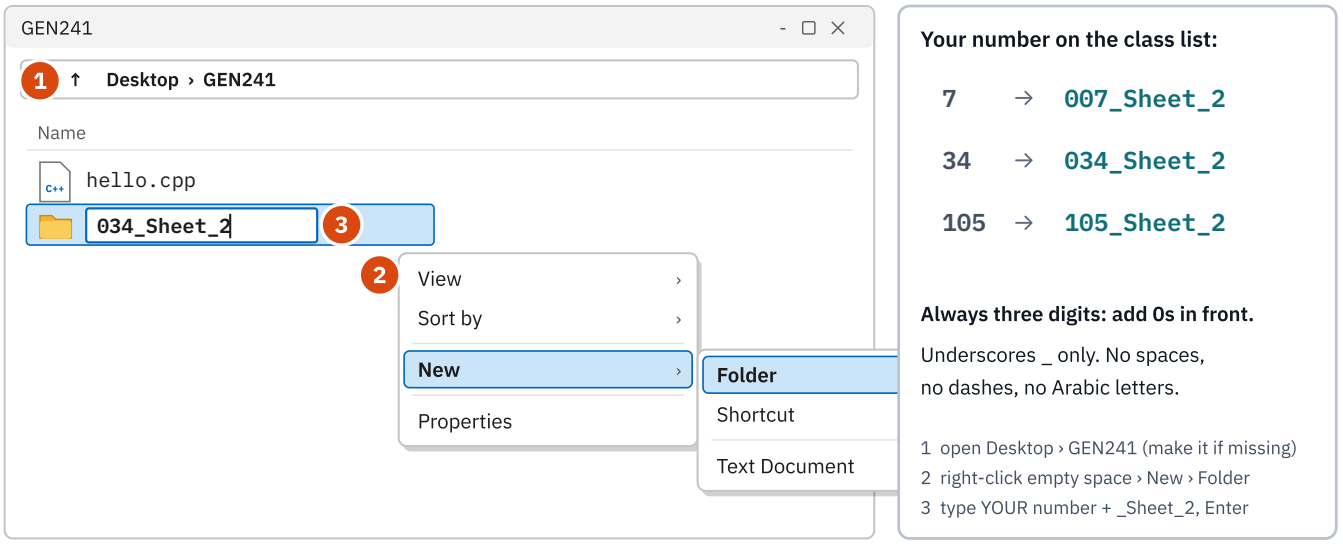
Follows the rules. A correct sheet folder name.

6.4

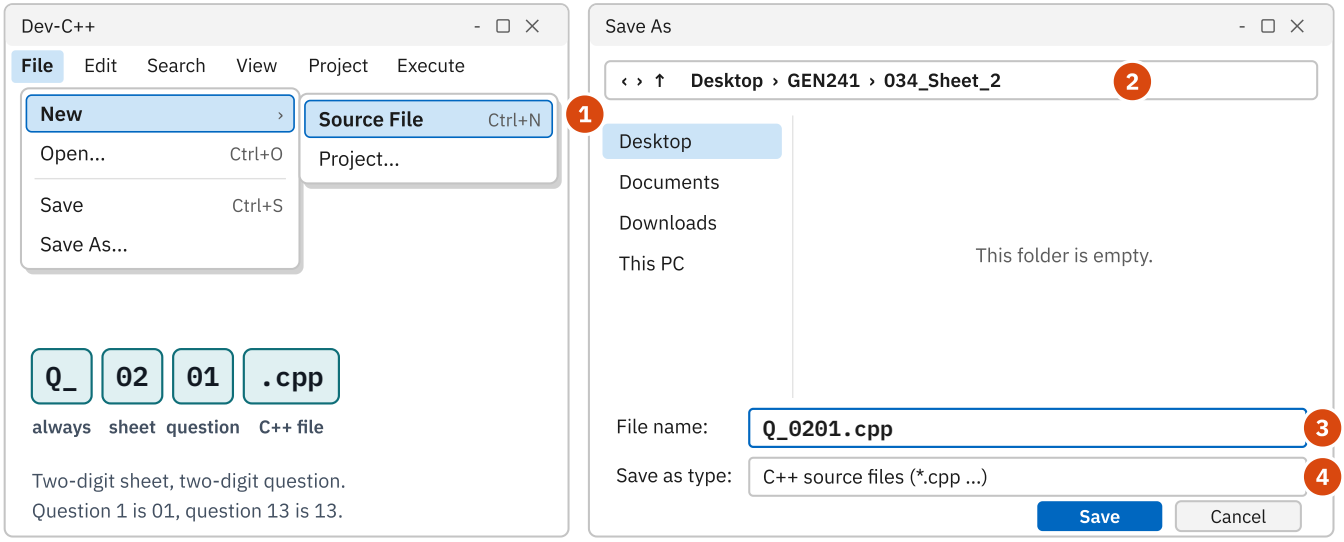
On a lab PC, step by step

The pictures are annotated illustrations of Windows 10. Windows 11 has the menu path given below.

1. Open `Desktop › GEN241` - the folder from week 1. If it is not there, make it: right-click the desktop, *New › Folder*, type `GEN241`.
2. Inside it, right-click an empty space, *New › Folder*, and type your folder name, e.g. `034_Sheet_2`. Press Enter.
3. Make sure *File name extensions* are switched on (Windows 10: *View › File name extensions*; Windows 11: *View › Show › File name extensions*), or `Q_0201.cpp.txt` looks like `Q_0201.cpp`.
4. In Dev-C++: *File › New › Source File*, then save it **before typing**: *File › Save*, go into your sheet folder, type `Q_0201.cpp`, and check that *Save as type* says C++.



The folder goes inside GEN241, named with your own number. Not 034 unless you are 034 on the list.



Save before you type. The file then lives in the sheet folder from its first line, with the name the professor specifies.

بالمصري

الرقم في الاسم هو رقمك في الكشف، دايمًا 3 أرقام: 7 تبقى 007. وسؤال 1 في شيت 2 يبقى Q_0201.cpp.
مفيش عربي ولا مسافات ولا شرطة.

WHAT COULD FOOL YOU

- **The wrong number.** Your university ID, the PC's number or your partner's number all make a valid-looking name that is still wrong.
- **Hidden extensions.** With extensions hidden, Windows shows `Q_0201.cpp.txt` as `Q_0201.cpp`. Switch them on once and it stays on.
- **Working inside Downloads.** A file opened from a WhatsApp download saves back to Downloads. Save it into your sheet folder first.

CHECK YOURSELF

You are number 105 on the list. Which is the file for sheet 2, question 8?

A Q_028.cpp

B 105_Q_0208.cpp

C Q_0208.cpp

D Q-0208.cpp

Back to the sheet: make `XXX_Sheet_2` now, and save every question into it as you go. Sheet 1 is handed in the same way: `XXX_Sheet_1` with `Q_0101.cpp` to `Q_0112.cpp`.

CHAPTER 7

Build and inspect the final archive

Package the latest files and check the actual archive before uploading.

Read › **Decide** › **Check** › Name › Hand in

7.1 Complete every question file

Your final Sheet 2 folder needs `Q_0201.cpp` through `Q_0218.cpp`. Week 3 continues Sheet 2, so do not change the prefix to Q_03. Sheet 1 uses Q_0101.cpp through Q_0112.cpp in its own XXX_Sheet_1 folder.

7.2 Submission timing

The message says two weeks. **18 October 2026 is inferred from 4 October**; check the professor's final announcement for the exact time and upload link. The naming PDF accepts either ZIP or RAR. These pictures use ZIP.

7.3

Before you zip it

Ticks are kept in this browser only. Zip at the very end: a zip is a copy of the folder at that moment, and editing a `.cpp` afterwards does not change it.

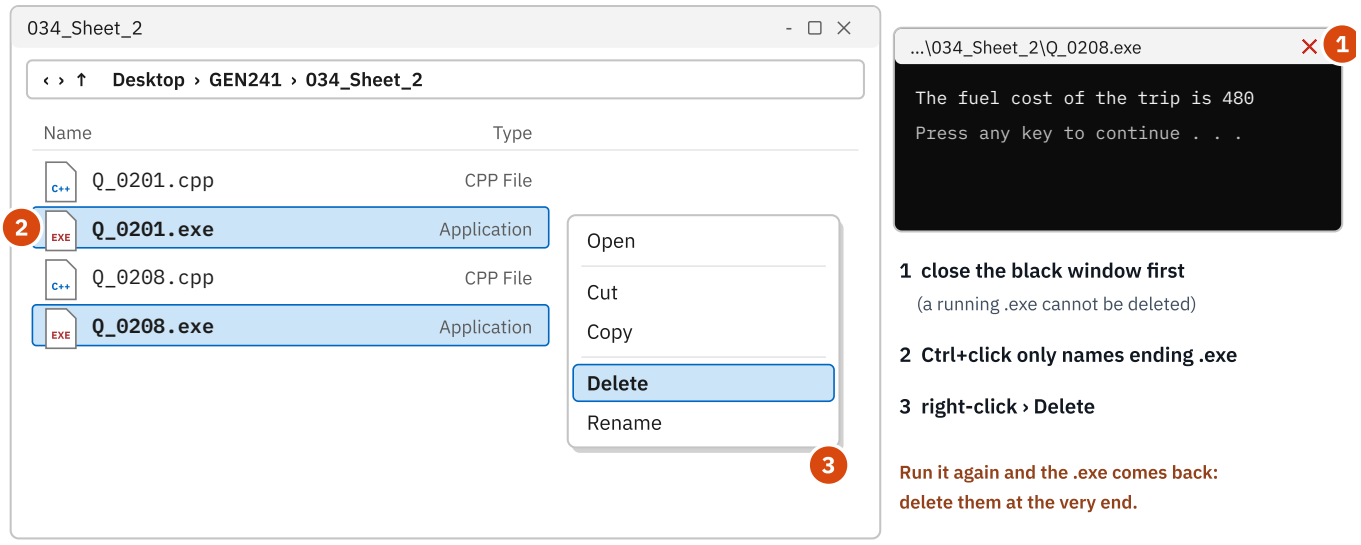
- [ ] **The folder is your number, three digits, _Sheet_2**
  034_Sheet_2, inside Desktop › GEN241. Chapter 6's builder makes it for you.
- [ ] **Every file is Q_02 + two digits + .cpp**
  All 18: Q_0201.cpp through Q_0218.cpp. Extensions switched on, so .cpp.txt cannot hide.
- [ ] **Every file compiles after your last edit**
  Compile each one again. "It worked yesterday" is not a test.
- [ ] **Every program passes its normal and boundary cases**
  Use the tests in chapters 2-5 and last week's guide.
- [ ] **Every division checked on its own**
  Both sides whole numbers? Did you mean that?
- [ ] **The .exe files are deleted**
  Close the black window first. Delete only names ending .exe.
- [ ] **The zip holds one folder, named like the zip**
  Zip from GEN241, not from inside the folder. Open the zip and look.

7.4

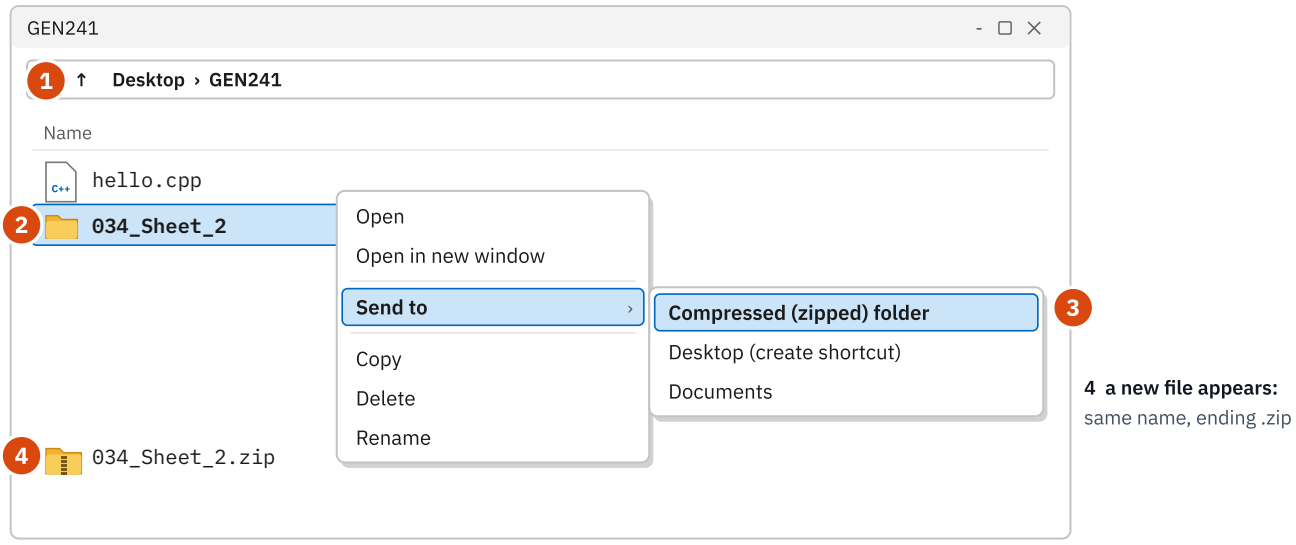
Packaging on a lab PC, picture by picture

Windows 11: right-click the folder, choose *Show more options › Send to › Compressed (zipped) folder*. Windows 10: *Send to › Compressed (zipped) folder*. If an older archive has the same name, move it aside or remove that old archive only. Keep your working folder.

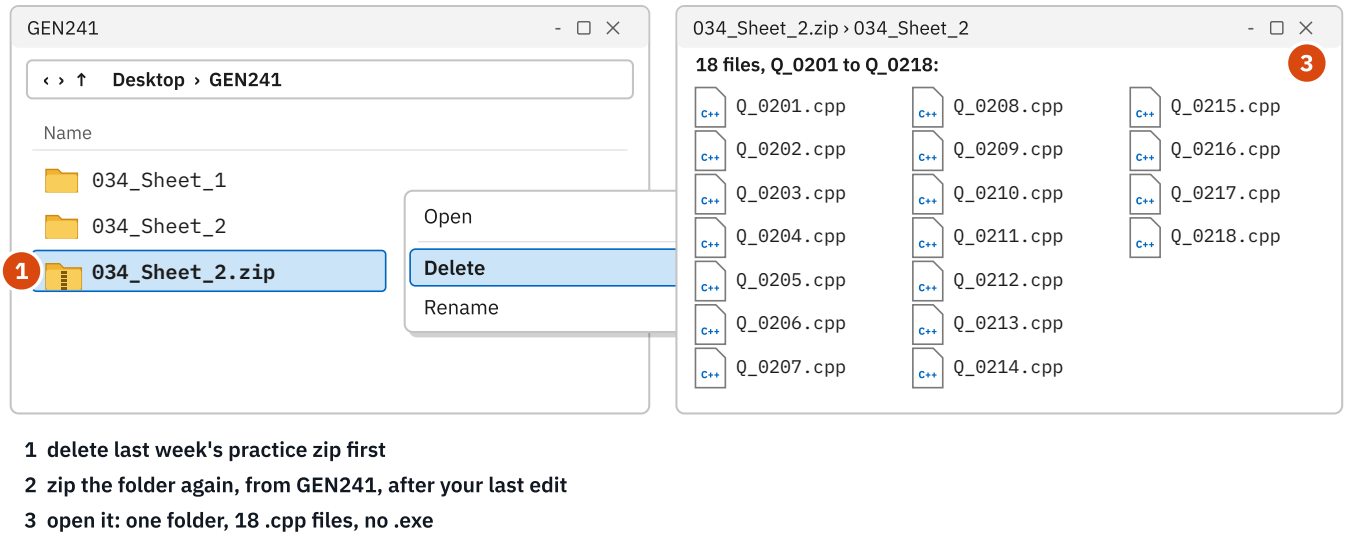
Do this once, at the very end, after the last test run. The same three moves package sheet 1 into `XXX_Sheet_1`.



Close the black window first. Windows will not delete an `.exe` that is still running. Delete only the names ending in `.exe`.



Zip the folder, from one level up. Zipping the files inside it gives a zip with no folder in it.



1 delete last week's practice zip first
2 zip the folder again, from GEN241, after your last edit
3 open it: one folder, 18 .cpp files, no .exe

The final archive contains one correctly named folder and all 18 source files.

After upload, check the website's confirmation and keep your working folder and submitted archive. Ticking this page does not upload anything.

Windows menu paths checked against Microsoft: file extensions and Microsoft: ZIP files. Pictures are explanatory illustrations, not captures from your lab PC.

Sources and checks

- Prof. Mahmoud M. Khater, Sheet 2 pp. 1-4, supplied reference code, and submission-rules.pdf. Read 5 October 2026.
- Lecture GEN_241_03 pp. 10-11 introduces selection and Boolean operators. No Lecture 4 file was supplied. Syntax explanations also use Liang, Introduction to Programming with C++, 3rd ed., §§3.3-3.6 and §3.10.
- Q18 labels reproduce the sheet's table for this programming exercise. The cable and current examples are additional teaching exercises.
- Examples compiled with Apple clang in C++98 mode, with checks recorded in CHECKED.md. Browser calculations were compared against compiled results.

Read on your own after the lab. Use your own class-list number and your own tested code.